



2950 Niles Road, St. Joseph, MI 49085-9659, USA
269.429.0300 fax 269.429.3852 hq@asabe.org www.asabe.org

An ASABE Meeting Presentation

DOI: <https://doi.org/10.13031/aim.202200175>

Paper Number: 2200175

Assessment of projected change in Intensity-duration-frequency (IDF) curves for Southeastern, United States using Artificial Neural Networks.

Bijoychandra S. Takhellambam¹, Puneet Srivastava², Jasmeet Lamba¹, Wenpeng Zhao³, Hemendra Kumar⁴, Di Tian¹

¹Auburn University, ²University of Maryland, ³Tokyo Institute of Technology, ⁴The Ohio State University.

**Written for presentation at the
2022 ASABE Annual International Meeting
Sponsored by ASABE
Houston, Texas
July 17–20, 2022**

ABSTRACT. Anthropogenic activities have increased greenhouse gas emissions in the atmosphere. The resulting changing climate is expected to affect the rainfall characteristic mostly amount, intensity, and frequency. The rainfall Intensity Duration Frequency (IDF) curves should be updated for the projected climatic conditions. Therefore, the objective of our study is to develop future projected IDF curves using the HADGEM model under the RCP 8.5 scenario over the southeast United States. We assess a feed forward back propagation Artificial Neural Network (ANN) model for downscaling 1-hour rainfall to sub-hour monthly maximum rainfall. The Nash Sutcliffe Efficiency was found in the range of 0.67 to 0.86. Moreover, the results from future projected rainfall IDF curves for a station located at Houston, Mississippi show increasing extreme rainfall event with 30-min and 45-min duration in range of 1% to 6% with a return period of 5, 10, 25, 50, and 100 years. However, 15-min duration show decreasing extreme rainfall depth in range of 2% to 4%. The spatial variation of extreme rainfall depth found both Gulf-Atlantic coast and the Appalachian Mountains are likely to receive greater rainfall than other parts of the southeast United States.

Keywords. ANN, Climate Change, GEV, stochastic, temporal disaggregation, Frequency analysis, IDF curves.

The authors are solely responsible for the content of this meeting presentation. The presentation does not necessarily reflect the official position of the American Society of Agricultural and Biological Engineers (ASABE), and its printing and distribution does not constitute an endorsement of views which may be expressed. Meeting presentations are not subject to the formal peer review process by ASABE editorial committees; therefore, they are not to be presented as refereed publications. Publish your paper in our journal after successfully completing the peer review process. See www.asabe.org/JournalSubmission for details. Citation of this work should state that it is from an ASABE meeting paper. EXAMPLE: Author's Last Name, Initials. 2022. Title of presentation. ASABE Paper No. ---. St. Joseph, MI.: ASABE. For information about securing permission to reprint or reproduce a meeting presentation, please contact ASABE at www.asabe.org/copyright (2950 Niles Road, St. Joseph, MI 49085-9659 USA).

INTRODUCTION

According to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC), anthropogenic activities have most likely contributed to the changing global water cycle (IPCC, 2013). One of the main reasons is increasing the atmosphere's water holding capacity, posing a serious threat to global rainfall (Trenberth, 2011). The existing hydraulic structures are designed based on the stationarity of the rainfall. However, rainfall characteristics such as amount, frequency and duration are projected to change with the climate change and climate variability. Moreover, the southeast United States has the greatest potential of changing rainfall characteristics under the changing climate (McGehee & Srivastava, 2018; Takhellambam et al., 2022a, 2022b). This necessitates to updates the IDF curves with the changing climate scenarios.

Previous studies have quantified the future projected rainfall Intensity Duration Frequency (IDF) curves. Mirhosseini et al. (2013) used 3-hour rainfall datasets to downscale to 15-min rainfall using a modified stochastic model. Using this dataset, the future projected IDF curves were developed for Alabama, United States. However, this downscaled rainfall intensities are under-predicted as compared to the observed rainfall (Mirhosseini et al., 2014; Takhellambam et al., 2022b). Mirhosseini et al. (2014) further improved the underpredicted rainfall intensities using a feed forward back propagation ANN model. Furthermore, this method was improved by Zhao et al. (2021) through a) more computationally efficient approach and b) better accuracy for estimation of monthly maximum rainfall. However, Zhao et al. (2021) used over two stations and downscaled daily to hourly maximum rainfall. Therefore, our study focused on the assessment applicability of ANN method to sub-hour approach. The objectives of our study are 1) downscaling of future projected hourly rainfall to sub-hour monthly maximum rainfall under RCP8.5 scenario using Artificial Neural Network, and 2) quantify the future projected IDF curves using 15-, 30-, and 45-min with a return period of 2, 5, 10, 25, 50, and 100 years over the southeast United States.

MATERIAL AND METHODS

DESCRIPTION OF STUDY AREA AND DATA

Fig.1 shows study area that includes eleven states of southeast United States- Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, and Virginia. The region received average annual rainfall that ranges from 1000 to 1500 mm as shown in Fig .1 (Kumar et al., 2021; 2022a; 2022b;Takhellambam et al., 2021). Higher rainfall amounts are received in the Gulf-Atlantic coast and Appalachian Mountain region and rainfall amount decreases away from these regions (Ingram et al., 2013).

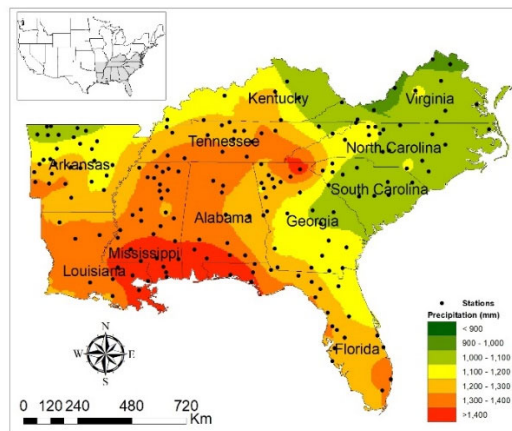


Figure 1. Map showing 187 rainfall stations that cover the entire 11 states of southeast United States.

15 Minute rainfall data (DSI-3260) archived at National Climate Data Center is obtained for 528 stations over 1970-213 covering the entire southeast United States (NOAA et al., 2014). Further, we used a screening method known as 20.11 to filter the bad rainfall stations, eventually we found 187 stations that satisfied the conditions (McGehee & Srivastava, 2018; Takhellambam et al., 2022a, 2022b). Moreover, we obtained hourly future projected rainfall for both the historical (1970-2010) and future (2030-59) period from North American-Coordinated Downscaling Experiment. We used Hadley Centre Global Environment Model version 2 Earth system model and Weather Research and Forecasting (herein we will use HADGEM) datasets, which is simulated with forcing Coupled Model Intercomparison Project Phase 5 (CMIP5) under representative concentration pathway (RCP) 8.5 and has a spatial resolution of 50 km (Mearns et al., 2009).

TEMPORAL DOWNSCALING OF RAINFALL USING ANN AND IDF CURVES

The climate models are associated with significant bias due to the systematic model errors of simplified representation of nature, discretization and spatial averaging within the rids (Teutschbein & Seibert, 2012). This will eventually make more complicated when using the climate models in the hydrological application. Therefore, we used quantile delta mapping method as discussed in Takhellambam et al. (2022b).

Further, our study used a feed forward back propagation of ANN model, which is proposed by Zhao et al. (2021). We assessed the performance of the model for downscaling sub-hour (15-, 30-, and 45-min) monthly maximum rainfall from 1-hour. In this method, we developed 187 models using 70% and 30% of rainfall data for training and testing, respectively. The input data parameters are concurrent hourly data and both preceding and succeeding hourly rainfall for a given 15-min data. Moreover, the performance evaluation is assessed using Nash Sutcliffe efficiency (NSE), Pearson correlation coefficient (r), and root mean square error (RMSE) (Mirhosseini et al., 2014; Takhellambam, 2016).

GENERATION OF IDF CURVES

Mirhosseini et al. (2013, 2014) suggest to use Generalized Extreme Value (GEV) distribution for developing IDF curves in the study area region. The generalized procedure to develop IDF curves is provided below.

- a) Select the annual maximum rainfalls for a given duration.
- b) Using the GEV distribution, calculate the rainfall depth for each return period of 2, 5, 10, 25, 50, and 100 years.

$$F(y) = \begin{cases} \exp \left\{ - \left[1 + \epsilon \left(\frac{y - \mu}{\sigma} \right)_+^{-1/\epsilon} \right] \right\}, & \text{if } \epsilon \neq 0 \\ \exp \left\{ - \exp \left(- \frac{y - \mu}{\sigma} \right)_+ \right\}, & \text{if } \epsilon \rightarrow 0 \end{cases}$$

where, $y_+ = \max(y, 0)$ with '+' positive, μ , σ , and ϵ are location, scale, and shape parameters respectively.

- c) Plot the rainfall depths against duration for different return periods.

RESULTS AND DISCUSSION

PERFORMANCE OF ANN

The bias corrected hourly rainfall are downscaled to 15-, 30-, and 45-min monthly maximum rainfall using the feed forward back propagation ANN model. The assessment of ANN performance was carried out by aggregating the observed 15-min rainfall to 1-hour rainfall. The 1-hour rainfall data is downscaled to 15-min monthly maximum rainfall. Similarly, we downscaled for the remaining 30-min and 45-min rainfall data.

Table 1 shows the average r and RMSE for both the training and testing period. The r value varies from 0.84 to 0.95 and 0.77 to 0.93 in training and model testing, respectively. The performance of the model improves with larger temporal scales

of 30-min and 45-min as compared to that of 15-min. In addition, we found the efficiency of the model for generating 15-, 30-, and 45-min monthly maximum rainfall with a NSE value of 0.67, 0.80 and 0.86, respectively. The performance is found satisfactory as NSE greater than 0.6 (Mirhosseini et al., 2014).

Table 1: Performance of downscaling of sub-hour monthly maximum rainfall from 1-hour using ANN model.

Duration	Training	Testing	Training	Testing
	r		RMSE	
15-min	0.84	0.77	0.37	0.48
30-min	0.92	0.90	0.40	0.34
45-min	0.95	0.93	0.23	0.28

IDF CURVES

Fig.2 shows the spatial distribution of extreme rainfall depth using HADGEM model. This result shows that the extreme rainfall depth mostly occurred in the Gulf-Atlantic coast and Appalachian Mountain, which is consistent with that of Takhellambam et al. (2022a, 2022b). This was likely due to convective and orographic effects in the Gulf-Atlantic coast and Appalachian Mountains, respectively. We also found that 15-min rainfall has greater intensity as compared to the larger temporal scale of 30 and 45-min. This is mainly due to the smoothing of rainfall intensities at larger scales. For instance, the maximum rainfall intensity of a storm event occurred during the first 15-min (McGehee & Srivastava, 2018).

For better discussion, we randomly selected a station located in Houston, Mississippi (33.88°N, 88.99°W) as shown in Fig. 3 and Table 2. We found that the observed and ANN model's extreme rainfall depth ranges from 39 to 171 mm/hr and 38 to 167 mm/hr, respectively. Under different rainfall durations, we found mixed results. In all the return periods, extreme rainfall depth with 15-min decreases in the range of 2% to 4%. The remaining rainfall duration of 30-min and 45-min show increasing in all of the return periods except 2-years. The increasing rate of extreme rainfall depth ranges from 1% to 6%.

Table 2: Extreme rainfall depth for 15-, 30-, and 45-min with return periods of 2, 5, 10, 25, 50, and 100 years for a station located at Houston, Mississippi (33.88°N, -88.99°E).

Duration	Return Period (years)											
	2		5		10		25		50		100	
	OBS	ANN	OBS	ANN	OBS	ANN	OBS	ANN	OBS	ANN	OBS	ANN
15-min	83	79	106	103	122	118	142	138	156	152	171	167
30-min	55	53	70	71	81	83	94	98	104	109	113	119
45-min	39	38	51	52	59	61	69	72	76	80	84	89

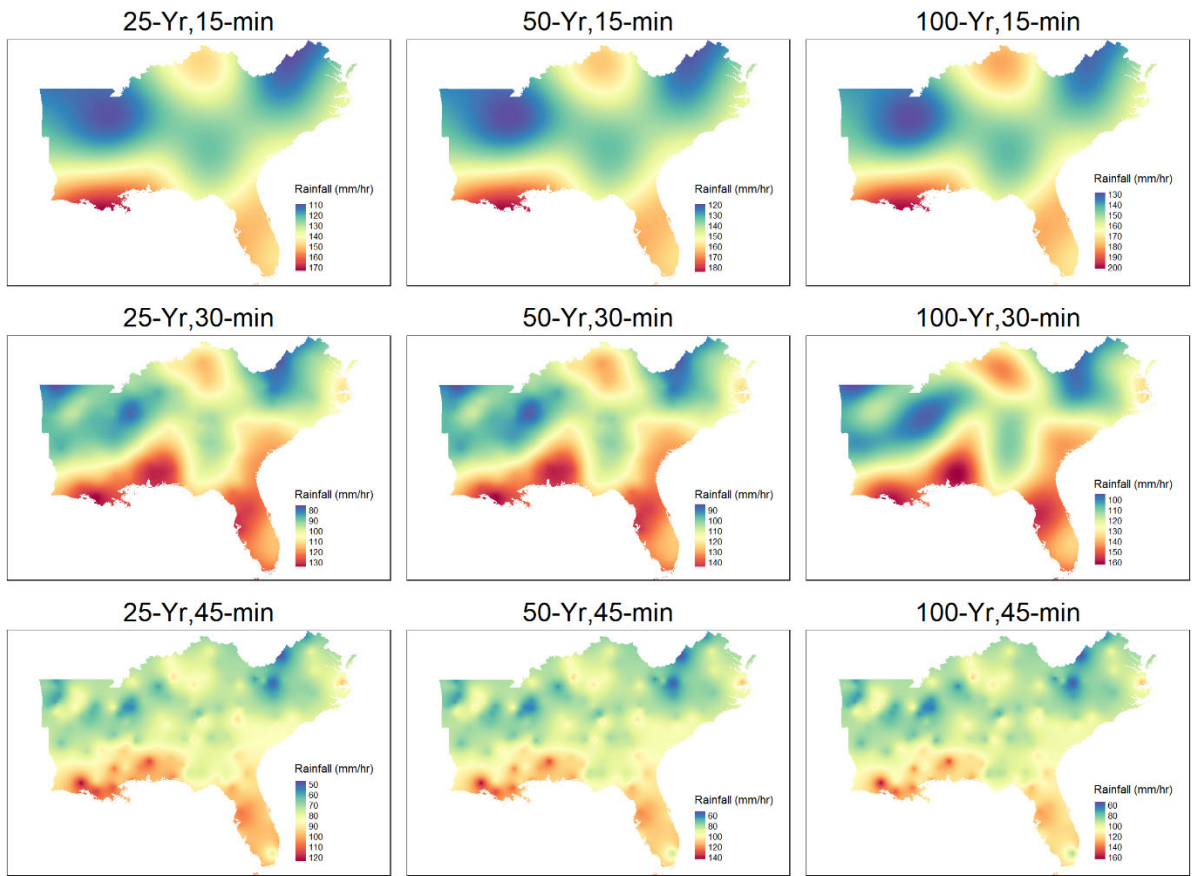


Fig. 2: Spatial variation of extreme rainfall depth for 15-, 30-, and 45-min with return period of 25-, 50-, and 100-year under HADGEM with RCP8.5 scenario.

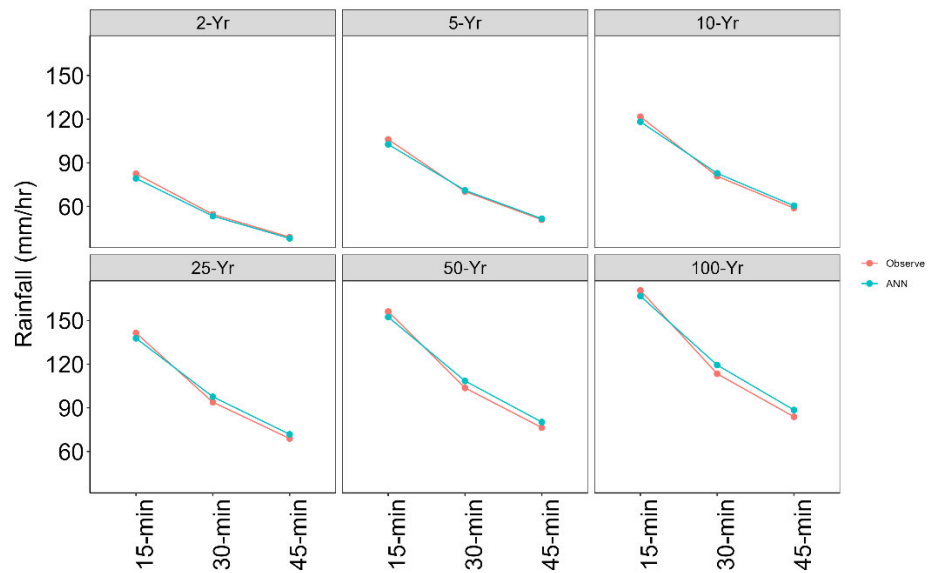


Fig 3: IDF curves for a randomly selected station located at Houston, Mississippi (33.88°N, 88.99°W).

The generated future projected IDF curves are associated with various uncertainties of observed and model rainfall datasets. The observed rainfall datasets are underpredicted with systematic errors of wind effects and evaporation (Zhao et al., 2021, 2022). Moreover, climate models have different initial and boundary conditions, and mechanisms for generation of rainfall. Therefore, every models are associated with its limitation (Mirhosseini et al., 2014).

CONCLUSIONS

We assessed a feed forward back propagation ANN model for downscaling hourly rainfall to monthly maximum sub-hour (15-,30-, and 45-min) rainfall depth under changing climate using HADGEM model with RCP 8.5 scenario. The statistical finding showed a satisfactory performance with NSE in the range of 0.67 to 0.86. The spatial variation of extreme rainfall depths shows an increasing extreme rainfall depth over the southeast United States. The Gulf-Atlantic coast and Appalachian Mountains are likely to receive more rainfall than other parts of the region. Furthermore, the comparison of extreme rainfall depth for a station located in Houston, Mississippi showed a mixed result. With 15-min rainfall duration, the future projected extreme rainfall depths for return periods of 2, 5, 10, 25, 50, and 100-year are found to decrease with values ranging from 2% to 4% as compared to the observed rainfall. While, 30-min and 45-min show an increasing rainfall depth with return periods of 5, 10, 25, 50, and 100 years in the range of 1% to 6%.

ACKNOWLEDGEMENTS

This work was completed in part with resources provided by the Auburn University Hopper Cluster. We are grateful for the support of USDA-NIFA Hatch Project (ALA014-1-19052), and the Alabama Agricultural Experiment Station funded this study.

REFERENCES

- Ingram, K., Dow, K., Carter, L., & Anderson, J. (2013). *Climate of the southeast united states: Variability, change, impacts, and vulnerability*. Island Press. <https://link.springer.com/book/10.5822/978-1-61091-509-0>
- IPCC. (2013). *Summary for Policymakers. In: Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press.
- Kumar, H., Srivastava, P., Lamba, J., Diamantopoulos, E., Ortiz, B., Morata, G., Takhellambam, B., & Bondesan, L. (2022a). Site-specific irrigation scheduling using one-layer soil hydraulic properties and inverse modeling. *Agricultural Water Management*, 273, 107877. <https://doi.org/10.1016/j.agwat.2022.107877>
- Kumar, H., Srivastava, P., Lamba, J., Ortiz, B. V., Way, T. R., Sangha, L., Takhellambam, B. S., Morata, G., & Molinari, R. (2022b). Within-Field Variability in Nutrients for Site-Specific Agricultural Management in Irrigated Cornfield. *Journal of Agricultural Safety and Health*, 65(4), 865–880. <https://doi.org/10.13031/ja.15042>
- Kumar, H., Srivastava, P., Ortiz, B. V., Morata, G., Takhellambam, B. S., Lamba, J., & Bondesan, L. (2021). Field-Scale spatial and temporal Soil water variability in irrigated Ccroplands. *Transactions of the ASABE*, 64(4), 1277–1294. <https://doi.org/10.13031/trans.14335>
- McGehee, R., & Srivastava, P. (2018). Benchmarking reliable erosion indices from quarter-hour station data for climate studies in the southeastern United States. *Journal of Soil and Water Conservation*, 73(4), 363–376. <https://doi.org/10.2489/jswc.73.4.363>
- Mearns, L. O., Gutowski, W. J., Jones, R., L.Y, L., S., M., A.M.B, N., & Y., Q. (2009). *A regional climate change assessment program for North America*. 90, 311–312.
- Mirhosseini, G., Srivastava, P., & Fang, X. (2014). Developing Rainfall Intensity-Duration-Frequency Curves for Alabama under Future Climate Scenarios Using Artificial Neural Networks. *Journal of Hydrologic Engineering*, 19(11), 04014022. [https://doi.org/10.1061/\(ASCE\)HE.1943-5584.0000962](https://doi.org/10.1061/(ASCE)HE.1943-5584.0000962)
- Mirhosseini, G., Srivastava, P., & Stefanova, L. (2013). The impact of climate change on rainfall Intensity–Duration–Frequency (IDF) curves in Alabama. *Regional Environmental Change*, 13(1), 25–33.

<https://doi.org/10.1007/s10113-012-0375-5>

- NOAA, Administration, National Centers for Environmental, & Information). (2014). *US 15 Minute Precipitation Data, Version 1.0. 1970-2010*. Boulder, CO: NOAA National Centers for Environmental Information.
- Takhellambam, B. S. (2016). *Flood discharge estimation only using at-site stage information*. [India Institute of Technology Roorkee]. <http://localhost:8081/xmlui/handle/123456789/14179>
- Takhellambam, B. S., Srivastava, P., Lamba, J., McGehee, R., & Kumar, H. (2021, December 13). *Potential changes in rainfall erosivity under climate change in southeastern United States*. AGU Fall Meeting 2021. <https://agu.confex.com/agu/fm21/meetingapp.cgi/Paper/989740>
- Takhellambam, B. S., Srivastava, P., Lamba, J., McGehee, R. P., Kumar, H., & Tian, D. (2022a). *Projected Rainfall Erosivity Under Climate Change in the Southeastern United States*. 1. <https://doi.org/10.13031/aim.202200176>
- Takhellambam, B. S., Srivastava, P., Lamba, J., McGehee, R. P., Kumar, H., & Tian, D. (2022b). Temporal disaggregation of hourly precipitation under changing climate over the Southeast United States. *Scientific Data*, 9(1), 211. <https://doi.org/10.1038/s41597-022-01304-7>
- Teutschbein, C., & Seibert, J. (2012). Bias correction of regional climate model simulations for hydrological climate-change impact studies: Review and evaluation of different methods. *Journal of Hydrology*, 456–457, 12–29. <https://doi.org/10.1016/j.jhydrol.2012.05.052>
- Trenberth, K. E. (2011). Changes in precipitation with climate change. *Climate Research*, 47(1–2), 123–138. <https://doi.org/10.3354/cr00953>
- Zhao, W., Abhishek, Kinouchi, T., Ang, R., & Zhuang, Q. (2022). A framework for quantifying climate-informed heavy rainfall change: Implications for adaptation strategies. *Science of The Total Environment*, 835, 155553. <https://doi.org/10.1016/j.scitotenv.2022.155553>
- Zhao, W., Kinouchi, T., & Nguyen, H. Q. (2021). A framework for projecting future intensity-duration-frequency (IDF) curves based on CORDEX Southeast Asia multi-model simulations: An application for two cities in Southern Vietnam. *Journal of Hydrology*, 598, 126461. <https://doi.org/10.1016/j.jhydrol.2021.126461>